

Team MoBI Fall 2024 Final Report

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Project Sprint Goals

Fall 2024

Sprint 0 (Sept 12 – Oct 6): Submit HC and replicate ridge regression results across all systems

Sprint 1 (Oct 7 - Nov 3): # of time points vs. R^2 , finalize HC and pitch deck, identify feature importance algorithms, replicate random forest regression results

Sprint 2 (Oct 21 - Dec 15): Normalization and application of ridge and random forest regression, LOOCV implementation, Implement Sillyf

Spring 2025

Sprint 3 (Dec 16 - Mar 4): Transfer learning, implement feature importance, apply identified regression model, All $0.5 < R^2 < 1$

Sprint 4 (Mar 5 - May 5): Finalize and submit the manuscript for publication

Replication of Ridge Regression

Replicating across Systems

To ensure that all the teammates had their environment variables and the setup to run Python environments were identical, each teammate was tasked with running the standard code provided by the PhD student and comparing the error across each person's system. This was simply a manner of loading the dataset and running the code on all computers, as all random states were the same, resulting in no error across individual laptops. Further investigation of the variation of results as the random state changed revealed that the error was less than 1%.

Penalty Optimization and Comparison with Sklearn

To better understand the implementation of ridge regression and the effect of the associated penalty on the model's performance on raw and normalized data, ridge regression code was written from scratch. The penalty value that resulted in the best performance was determined to be 0.01, and this penalty was applied for all instances of ridge regression moving forward.

The Sklearn statistical learning package was then used to rewrite the ridge regression code to confirm the performance of the model written from scratch. Both sets of code yielded consistent results. This Sklearn code was then utilized, along with a penalty of 0.01, for all implementations of ridge regression moving forward.

Dependence of Model Performance on the Number of Data Points Used

Given the large volume of features and data streams, the team attempted to gain further insight into the performance of the ridge regression model as a function of the number of data streams. This involved implementing code that would take all rows, i.e. data points of a single test subject and split it into different volume fractions of test and training data accessing the least required amount of training and test data split. This led to the conclusion that a training-to-test data split of 0.4:0.6 would give reasonable and consistent results, as increasing this fraction has no computational benefit. The team also noticed that the R^2 score was <0 , which indicated that the model was not performing well for the data set and would require modifications. The

modifications were explored regarding available pre-processing methods, namely normalization, which are enumerated in further sections.

Replication of Random Forest Regression

Replicating across Systems and Multiple Trials

To ensure that the random forest model's performance was consistent and reproducible, each team member was tasked with running the same code on their individual system with multiple trials and comparing errors across systems. Replicating across systems was necessary to test if the model's performance was hardware-dependent or influenced by individual operating systems or Python environments. Multiple trials on the same system evaluated the stability of the model's output, which is an effective way to identify whether the inconsistencies were due to random seed initialization or inherent variability in the algorithm.

Errors encountered during Random Forest Regression

When replicating the code of the random forest, all individual laptops did not yield similar results, similar to what was observed with ridge regression. The results were also not consistent when multiple trials were conducted on the same laptop. While further looking at various possibilities for this deviation, it was suggested that using multithreading might have been the source of the error. This was eliminated by using a Linux system that ran the code to work on the same core with multiple iterations. This still showed consistency error between iterations, this was later resolved as one of the initializations of the random seed was not consistent hence leading to the variation between individual laptops and multiple iterations in a singular laptop.

Upon resolving the variation errors by explicitly calling the random seed prior to every trial of the random forest regression, the results gathered from the raw data were observed. Three sample fractions on a logarithmic scale from 1000 to the total number of data streams, different for each subject, were utilized and sample fractions above approximately 0.8 were observed to provide R^2 values between 0.8 and 1, indicating that the random forest regression model was a good fit for the data.

Normalization Preprocessing

To ensure fair contribution of all features to the Ridge regression model, normalization techniques were explored. Ridge regression, which incorporates an L_2 regularization term, often benefits from normalized data to stabilize the optimization process.

Three normalization techniques were applied to the dataset:

- Z-Score Normalization: Scaled each feature to have a mean of 0 and a standard deviation of 1
- Min-Max Scaling: Transformed features to lie within the [0, 1] range
- Robust Scaling: Used the interquartile range (IQR) for scaling, reducing the influence of outliers

Additionally, heatmaps were generated for the normalized data to visually assess the distribution of features across the dataset and to compare the effects of the different normalization techniques.

5-Fold Means Cross Validation with Ridge Regression

The performance of the ridge regression model with 5-fold cross-validation was evaluated across different sample fractions of the dataset. Given that feature scale impacts the weighting of the features upon the implementation of ridge regression, normalization was expected to result in improved prediction scores when applying ridge regression. However, the mean R^2 scores remained consistent across all methods, indicating that normalization did not significantly impact model performance. The model was applied for all 61 subjects, for all ~17,000 data streams per subject across all features were used to generate predictions for all three targets: position difference, stimulus position and stimulus speed.

5-Fold Means Cross Validation with Random Forest Regression

The performance of the random forest model with 5-fold means cross-validation was evaluated across different sample fractions of the dataset. Given that the feature scale does not have a significant impact on the performance of random forest regression, no significant change in prediction scores was expected upon the application of this model to the normalized data compared to the raw data. This hypothesis held true as the mean R^2 scores remained consistent across all methods, indicating that normalization did not significantly impact model performance. The model was applied for all 61 subjects, for all ~17,000 data streams per subject across all features were used to generate predictions for all three targets: position difference, stimulus position and stimulus speed. This approach contrasts with linear methods like Ridge Regression, offering a different perspective on the data analysis.

Leave-One-Out Cross Validation with Ridge Regression

While the previously implemented ridge regression involved 5-fold cross validation, both Ridge and Random Forest regression models were also implemented using Leave-One-Out cross validation. For ridge regression, this method was performed both within a subject and across subjects. Within a subject, for all 61 subjects, all ~17,000 data streams across all features were used to train the model for all three targets: position difference, stimulus position and stimulus speed. Across subjects, the model was trained on all data features, ~17000 data streams, across 60 subjects and tested using 1 subject, with this process repeated with testing occurring for each of the 61 subjects. While the mean absolute error values seemed promising with values < 0.05 , the R^2 values fell between -0.6 and 0.1, indicating a poor fit of the model and applied cross-validation. This model was implemented on all sets of normalized data to make comparisons between model performance with different preprocessing techniques applied.

Leave-One-Out Cross Validation with Random Forest Regression

For random forest regression, Leave-One-Out cross validation was applied across the subjects. Training consisted of processing all ~17,000-time steps for each of the 14 input data streams, along with the corresponding targets: position difference, stimulus position, and stimulus speed, for each subject. The model was trained on the data from 60 subjects and then tested on the one subject remaining. This process was repeated for three subjects, and the results were compared across their respective datasets, which were normalized using various methods: min-max, z-score, and robust scaling. The R^2 values consistently ranged between -3.0 and 0.1, indicating a poor model fit under the applied configurations.

Next Steps

- Implement transfer learning Sillyf algorithm for improved clarity and usability
- Apply feature importance algorithm to highlight key predictors
- Apply the selected regression model with Sillyf and feature importance
- Target R^2 values range between 0.5 and 1
- Write submission draft for manuscript
- Deliverables:
 - Refined model outputs
 - Feature importance analysis
 - Complete manuscript draft